

Membrane Circular

BSA, 2024

```
In[55]:= ClearAll["Global`*"]  
Needs["VariationalMethods`"]
```

Euler-Lagrange Equation

```
In[57]:= x = {r, ϕ, t};  
U = u[r, ϕ, t];
```

```
In[59]:= T = 1 / 2 * λ * D[U, t] ^ 2 * r;  
P = κ * (Sqrt[1 + D[U, r] ^ 2 + (D[U, ϕ] / r) ^ 2] - 1) * r;
```

```
In[61]:= L = T - P
```

Out[61]=

$$\frac{1}{2} r \lambda u^{(\theta, \theta, 1)}[r, \phi, t]^2 - r \kappa \left(-1 + \sqrt{1 + \frac{u^{(\theta, 1, \theta)}[r, \phi, t]^2}{r^2} + u^{(1, \theta, \theta)}[r, \phi, t]^2} \right)$$

```
In[62]:= LaEq = Simplify[EulerEquations[L, U, {r, ϕ, t}]]
```

```
Out[62]=
```

$$\left(r^2 \kappa u^{(1,0,0)}[r, \phi, t]^3 + r \kappa u^{(0,2,0)}[r, \phi, t] \left(1 + u^{(1,0,0)}[r, \phi, t]^2 \right) - r^3 \lambda u^{(0,0,2)}[r, \phi, t] u^{(1,0,0)}[r, \phi, t]^2 \right. \\ \left. \sqrt{1 + \frac{u^{(0,1,0)}[r, \phi, t]^2}{r^2}} + u^{(1,0,0)}[r, \phi, t]^2 + \kappa u^{(1,0,0)}[r, \phi, t] \left(r^2 + 2 u^{(0,1,0)}[r, \phi, t]^2 - 2 r u^{(0,1,0)}[r, \phi, t] u^{(1,1,0)}[r, \phi, t] \right) - \right. \\ \left. r \left(r^2 + u^{(0,1,0)}[r, \phi, t]^2 \right) \left(\lambda u^{(0,0,2)}[r, \phi, t] \sqrt{1 + \frac{u^{(0,1,0)}[r, \phi, t]^2}{r^2}} + u^{(1,0,0)}[r, \phi, t]^2 - \kappa u^{(2,0,0)}[r, \phi, t] \right) \right) / \\ \left(\sqrt{1 + \frac{u^{(0,1,0)}[r, \phi, t]^2}{r^2}} + u^{(1,0,0)}[r, \phi, t]^2 \left(u^{(0,1,0)}[r, \phi, t]^2 + r^2 \left(1 + u^{(1,0,0)}[r, \phi, t]^2 \right) \right) \right) = 0$$

Linearized Equation

```
In[63]:= Vars = Flatten[Table[D[U, {x[[i]], j}], {i, 3}, {j, 2}]]
```

```
Out[63]=
```

$$\{u^{(1,0,0)}[r, \phi, t], u^{(2,0,0)}[r, \phi, t], u^{(0,1,0)}[r, \phi, t], u^{(0,2,0)}[r, \phi, t], u^{(0,0,1)}[r, \phi, t], u^{(0,0,2)}[r, \phi, t]\}$$

```
In[64]:= LaGrad = Simplify[Table[D[LaEq[[1]], Vars[[i]] /. Thread[Vars → 0], {i, Length[Vars]}]]
```

```
Out[64]=
```

$$\left\{ \kappa, r \kappa, 0, \frac{\kappa}{r}, 0, -r \lambda \right\}$$

```
In[74]:= LaEqLin = Expand[Simplify[LaGrad . Vars] / r] == 0
```

```
Out[74]=
```

$$-\lambda u^{(0,0,2)}[r, \phi, t] + \frac{\kappa u^{(0,2,0)}[r, \phi, t]}{r^2} + \frac{\kappa u^{(1,0,0)}[r, \phi, t]}{r} + \kappa u^{(2,0,0)}[r, \phi, t] = 0$$

Linear Non-Dimensionalization

In[75]:= $\xi = \{\rho, \phi, \tau\};$
 $V = v[\rho, \phi, \tau];$

In[77]:= **NonDimToVars** = $\{\rho \rightarrow r / R, \phi \rightarrow \phi, \tau \rightarrow t * \Omega\};$
VarsToNonDim = $\{r \rightarrow \rho * R, \phi \rightarrow \phi, t \rightarrow \tau / \Omega\};$

In[79]:= **UToNonDim** = Flatten[Table[D[U, {x[[i]], j}] → (D[R * V /. NonDimToVars, {x[[i]], j}] /. VarsToNonDim), {i, 3}, {j, 2}]]

Out[79]=
$$\left\{ u^{(1,0,0)}[r, \phi, t] \rightarrow v^{(1,0,0)}[\rho, \phi, \tau], u^{(2,0,0)}[r, \phi, t] \rightarrow \frac{v^{(2,0,0)}[\rho, \phi, \tau]}{R}, u^{(0,1,0)}[r, \phi, t] \rightarrow R v^{(0,1,0)}[\rho, \phi, \tau], \right.$$

$$\left. u^{(0,2,0)}[r, \phi, t] \rightarrow R v^{(0,2,0)}[\rho, \phi, \tau], u^{(0,0,1)}[r, \phi, t] \rightarrow R \Omega v^{(0,0,1)}[\rho, \phi, \tau], u^{(0,0,2)}[r, \phi, t] \rightarrow R \Omega^2 v^{(0,0,2)}[\rho, \phi, \tau] \right\}$$

In[80]:= **LaEqLin1** = LaEqLin /. UToNonDim /. VarsToNonDim

Out[80]=
$$-R \lambda \Omega^2 v^{(0,0,2)}[\rho, \phi, \tau] + \frac{\kappa v^{(0,2,0)}[\rho, \phi, \tau]}{R \rho^2} + \frac{\kappa v^{(1,0,0)}[\rho, \phi, \tau]}{R \rho} + \frac{\kappa v^{(2,0,0)}[\rho, \phi, \tau]}{R} == 0$$

In[81]:= **SolOmega** = $\{\Omega \rightarrow \text{Sqrt}[\kappa / \lambda] / R\}$

Out[81]=
$$\left\{ \Omega \rightarrow \frac{\sqrt{\frac{\kappa}{\lambda}}}{R} \right\}$$

```
In[82]:= LaEqLinNonDim = LaEqLin1 /. Sol0omega /. {κ → 1, R → 1}
```

Out[82]=

$$-\mathcal{U}^{(0,0,2)}[\rho, \phi, \tau] + \frac{\mathcal{U}^{(0,2,0)}[\rho, \phi, \tau]}{\rho^2} + \frac{\mathcal{U}^{(1,0,0)}[\rho, \phi, \tau]}{\rho} + \mathcal{U}^{(2,0,0)}[\rho, \phi, \tau] == 0$$